DAY 6 — 7-DAY DATA ANALYST PORTFOLIO PROJECT

Business Case Study

E-Commerce Sales Performance: Diagnosing Revenue Concentration and Order-Value Risk from Two Years of Transaction Data

Business Problem

Leadership needs to know whether current sales performance is healthy and sustainable, or whether it is quietly concentrated in a single region and a single low-value transaction pattern that puts future growth at risk.

By Alya Shandy | Dataset: cleaned_sales_data.csv (3,933 orders, Jan 2023 – Dec 2024)

THE DATASET

Two Years of E-Commerce Transactions

Every recommendation in this case study is grounded in a single, verified transaction dataset spanning two full years — enough history to separate real trends from monthly noise.

3,933	889	24	12
Orders	Unique Customers	Months (Jan 2023–Dec 2024)	Data Fields

Fields Captured per Order

- Order ID
- Order Date
- Customer ID / Name
- Region
- Category
- Product Name
- Quantity
- Unit Price
- Order Status
- Payment Method
- Revenue

DATA QUALITY

From Raw Export to Analysis-Ready Data

The raw export required real cleanup before any number in this report could be trusted. That work happened on Day 1 and every subsequent stage built on the result.

BEFORE — Raw Export

4,100

rows, with issues:

- ✖ Duplicate order records
- ✖ Missing values in key fields
- ✖ Revenue outliers from data-entry errors
- ✖ Inconsistent date & category formatting

→

AFTER — Analysis-Ready

3,933

clean rows, verified:

- ✔ 0 duplicate rows remaining
- ✔ 0 missing values across all 12 fields
- ✔ Outliers reviewed and corrected
- ✔ Standardized date & category formats

167 rows (4.1%) removed or corrected — every KPI in this report is calculated on the 3,933-row clean dataset.

KPI OVERVIEW

The Business at a Glance

$306,842

Total Revenue (Gross)

$252,228

Completed (82.2%)

3,933

Total Orders (Gross)

3,220

Completed (81.9%)

889

Total Customers

98.8% of orders are repeat

$78.02

Avg. Order Value (Gross)

$78.33

Completed

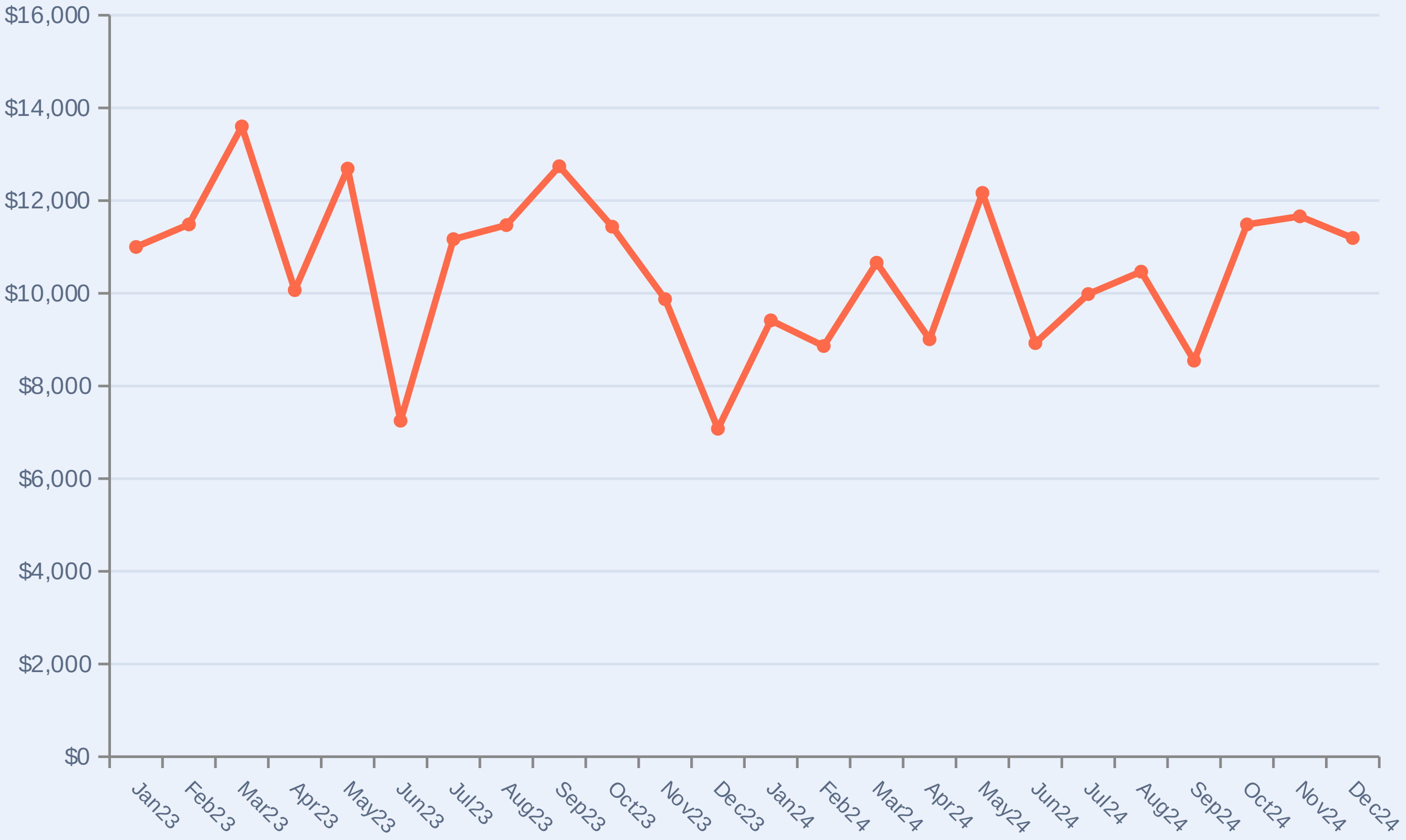
!

Gross vs. Completed — why the gap matters

17.8% of gross revenue ($54,614) sits in Cancelled, Returned, or Pending orders. Every chart later in this deck uses Completed-only figures unless labeled otherwise.

SALES PERFORMANCE

Revenue Grows Overall, but Swings Sharply Month to Month



+54.1%

Best month-over-month growth (Jul 2023)

-42.9%

Worst month-over-month drop (Jun 2023)

No clear seasonality

Swings recur across both years with no predictable calendar pattern

PRODUCT & CATEGORY

Electronics Leads — Books Lags Consistently

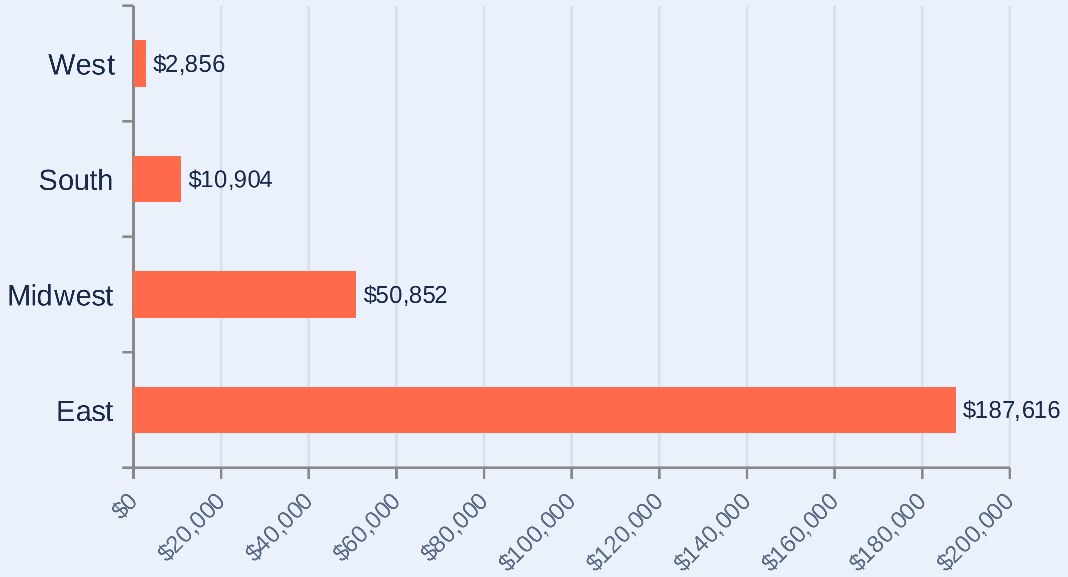
Top 3 Products by Revenue

1. Smartphone Case — $9,437
2. Webcam — $9,267
3. Power Bank — $8,509

25.3%

of total revenue comes from Electronics alone — more than double the next-closest category's contribution.

Loyal Customers, One Region, and a Value Gap



East region alone drives 74.4% of Completed revenue

98.8%

of orders come from repeat customers — retention is a genuine strength, but new-customer acquisition is nearly flat.

$78 vs $289

Average Order Value ($78.33) sits far below average revenue per customer ($288.59) — customers return often, but spend little each visit.

Price and quantity purchased are nearly uncorrelated (r = 0.02) — customers don't buy more units just because a product is cheaper.

KEY INSIGHTS

Five Findings That Matter Most

1. **Revenue is dangerously concentrated**

   74.4% of revenue depends on one region (East); West contributes just 1.1%.

2. **AOV is low relative to customer value**

   $78.33 per order vs. $288.59 per customer — the gap is a bundling opportunity, not a pricing problem.

3. **Retention is strong, acquisition is weak**

   98.8% of orders come from repeat customers — growth is not coming from new buyers.

4. **Electronics is the clear category leader**

   25.3% of revenue from one category, more than double the next-highest.

5. **Monthly performance is highly volatile**

   Growth swings from -42.9% to +54.1% month over month, with no seasonal pattern.

RECOMMENDATIONS

Four Actions to Reduce Risk and Grow Value

1 Diversify beyond the East region

Pilot targeted marketing and fulfillment investment in Midwest and South before West, using East's playbook as the template.

2 Launch bundling & cross-sell offers

Pair frequently-bought categories (e.g. Electronics accessories) to lift Average Order Value beyond $78.33 without discounting.

3 Invest in new-customer acquisition

Retention is already excellent (98.8% repeat); growth now depends on paid/referral channels that bring in first-time buyers.

4 Build a lightweight demand forecast

Use the -42.9% to +54.1% MoM swing pattern to plan inventory and staffing instead of reacting to each spike or drop.

CONCLUSION

A Healthy Business With Two Fixable Risks

Two years of clean transaction data tell a consistent story across Excel, SQL, Python, and Tableau: this business retains customers exceptionally well and has a clear category leader in Electronics — but revenue depends too heavily on one region, and order value stays low even as customers keep coming back. Both risks are addressable with the four recommendations on the prior slide, not structural problems requiring a business model change.

74.4%

revenue from one region

$78 vs $289

AOV vs. revenue per customer

98.8%

orders from repeat customers

Next: Day 7 — Portfolio Publication

Full analysis, dashboard, and documentation: github.com/aloutvbs/ecommerce-sales-analysis